Spectral–Spatial Fractal Residual Convolutional Neural Network With Data Balance Augmentation for Hyperspectral Classification

Xin Zhang, Yongcheng Wang, Ning Zhang, Dongdong Xu, Huiyuan Luo, Bo Chen, and Guangli Ben

***Abstract*— The development of deep learning has brought new prospects into the field of hyperspectral classification, and the classification ability of this method for the classification of hyperspectral images (HSIs) has been continuously improved. However, there are still some problems that must be solved; for example, the spectral–spatial features of HSIs are not effectively extracted, the labeled samples in the data set are limited, and the number of samples in different categories is imbalanced. To facilitate the progress of hyperspectral classification, a spectral–spatial fractal residual convolutional neural network with data balance augmentation is proposed here. In this network, a data balance augmentation approach that can solve the problems of limited labeled data and imbalanced categories is proposed. In addition, the spectral–spatial residual module is proposed to learn the spectral–spatial information and alleviate the problem of model degradation effectively. In addition, the spectral–spatial focal structure, which can guarantee the integrity of the information, is introduced. Moreover, the spectral–spatial dimensional transformation module, which can reduce the size and number of hyperspectral feature maps without losing the fine features, is presented. In particular, the proposed network has a strong ability to classify the categories that have a small number of samples and reaches the state-of-the-art level for three benchmark data sets.**

***Index Terms*— Convolutional neural network (CNN), hyperspectral classification, image processing, deep learning.**

Nomenclature and Abbreviations

AA	Average accuracy.
BN	Batch normalization.
CNN	Convolutional neural network.
DBN	Deep belief network.
HSIs	Hyperspectral images.
IP	Indian Pines.
IP_SSFRAN	SSFRAN with all the strategies on the IP data set.
IP_SSFRAN_wtBN	SSFRAN without BN on the IP data set.
IP_SSFRAN_wtD	SSFRAN without Dropout on the IP data set.
IP_SSFRAN_wtDT	The network that replaces the spectral–spatial dimensional transformation module in SSFRAN with the 3-D convolutional layer on the IP data set.
IP_SSFRAN_wtL2	SSFRAN without L2 regularization on the IP data set.
IP_SSFRN	SSFRAN without data balance augmentation on the IP data set.
IP_SSRAN	The network that only use one subpath of the spectral–spatial fractal structure in SSFRAN on the IP data set.
KNN	K-nearest neighbor.
KSC	Kennedy Space Center.
KSC_SSFRAN	SSFRAN with all the strategies on the KSC data set.
KSC_SSFRAN_wtBN	SSFRAN without BN on the KSC data set.
KSC_SSFRAN_wtD	SSFRAN without dropout on the KSC data set.
KSC_SSFRAN_wtDT	The network that replaces the spectral–spatial dimensional transformation module in SSFRAN with the 3-D convolutional layer on the KSC data set.
KSC_SSFRAN_wtL2	SSFRAN without L2 regularization on the KSC data set.
KSC_SSFRN	SSFRAN without data balance augmentation on the KSC data set.
KSC_SSRAN	The network that only use one subpath of the spectral–spatial fractal structure in SSFRAN on the KSC data set.
MNF	Minimum noise fraction.
OA	Overall accuracy.
PCA	Principal component analysis.
PU	Pavia University.

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PU_SSFRAN	SSFRAN with all the strategies on the PU data set.
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PU_SSFRAN_wtD	SSFRAN without dropout on the PU data set.
PU_SSFRAN_wtDT	The network that replaces the spectral–spatial dimensional transformation module in SSFRAN with the 3-D convolutional layer on the PU data set.
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PU_SSRAN	The network that only use one subpath of the spectral–spatial fractal structure in SSFRAN on the PU data set.
RF	Random forest.
RGB	Red, green, and blue.
RNN	Recurrent neural networks.
SAE	Stacked autoencoder.
SAM	Spectral angle mapper.
SSDTM1	The first spectral–spatial dimensional transformation in SSFRAN.
SSDTM2	The second spectral–spatial dimensional transformation in SSFRAN.
SSDTM3	The third spectral–spatial dimensional transformation in SSFRAN.
SSFRAN	The spectral–spatial fractal residual convolutional neural network with data balance augmentation.
SSFS1	The first spectral–spatial fractal structure in SSFRAN.
SSFS2	The second spectral–spatial fractal structure in SSFRAN.
SSFS3	The third spectral–spatial fractal structure in SSFRAN.
SVM	Support vector machine.

I. Introduction

As an important means of Earth observation, hyperspectral imaging can obtain HSIs with hundreds of continuous narrow bands using hyperspectral remote sensing sensors [1]. Different from traditional RGB images, HSIs present 3-D structures that contain both spectral and spatial features, thus making it possible to classify and identify relevant objects accurately [2]. More specifically, the classification of HSIs aims at predicting the category for each pixel, which is widely applied in mineral exploitation [3], environment monitoring [4], precision agriculture [5], and so on.

Although hyperspectral classification has made remarkable achievements, it still faces great challenges. There are two main problems.

1) The spatial distributions of the different objects in the HSIs are complex and there are mixed pixels [6]. In addition, there is a phenomenon that the same ground objects contain different spectral information and different ground objects contain the same spectral information [7].
2) The amount of labeled data is too small, and the cost of manually collecting and labeling new samples is too high [8]. Furthermore, the limited samples cannot balance the high-dimensional data, which is prone to the Hughes phenomenon; in other words, the classification performance decreases with increasing of spectral dimension [9].

For a long time, spectral-based methods have been adopted in the field of hyperspectral classification. According to whether high-level features are adopted or not, the spectral-based classification methods can be divided into traditional methods and methods using deep learning. In traditional spectral-based classification methods, the operations for dimension reduction, such as SAM [10], MNF [11], Jefferies–Matusita distance [12], and PCA [13], are usually used to extract spectral features. Then, the spectral features are sent into the classifiers, which include SVM [14], RF [15], KNN [16], and so on. With the development of technology, spectral-based methods that use deep learning for hyperspectral classification have gradually been proposed. They can automatically extract high-level semantic information from HSIs by simulating the recognition process of the brain, and they have achieved remarkable achievements in the classification of HSIs. Typical spectral-based methods that use deep learning include the RNN [17], SAE [18], 1-D CNN [19], and DBN [20]. The shortcoming of the spectral-based classification methods is that they ignore the spatial information of the HSIs.

To solve the complex spatial distribution of the HSIs, spectral–spatial classification methods have been developed gradually. They include two approaches: the methods that extract features jointly and the methods that extract features directly. Depending on the types of features adopted, the methods that extract spectral–spatial features jointly can be subdivided into two types: traditional methods and the deep learning-based methods. In the traditional classification methods of extracting spectral–spatial features jointly [21]–[23], the spectral features extracted by dimension reduction are combined with the spatial information obtained through attribute profiles [24], morphological profiles [25], sparse representations [26] and so on, which are used as the input of the classifier. The traditional spectral–spatial classification methods can improve the classification performance compared with the spectral-based classification methods, but they rely heavily on manual features that are extracted by experienced experts and have poor generalization ability. Fortunately, the spectral–spatial classification methods of extracting the spectral–spatial features jointly based on deep learning have been gradually developed. Hao *et al.* [27] proposed a two-stream architecture in which one stream encoded the spectral features by the stacked denoising autoencoder and the other steam extracted the spatial features by deep CNNs. Zhang *et al.* [28] introduced a dual-channel CNN, in which the spectral information was extracted by 1-D CNN and the spatial information was

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obtained by 2-D CNN. Zhao and Du [29] designed a spectral–spatial framework that used dimension reduction and 2-D CNN jointly for hyperspectral classification. However, these methods typically ignore the integrity of the HSIs, causing the loss of useful information. Under this circumstance, the superiority of the methods of extracting spectral–spatial features directly is shown. There are also two categories in the methods of extracting spectral–spatial features directly—the traditional methods and the deep learning-based methods. Concretely, the traditional classification methods of extracting the spectral–spatial features directly usually use the 3-D morphological profile [30], 3-D Gabor filters [31], and 3-D wavelet filters [32] to obtain the features. The classification methods of extracting spectral–spatial features directly based on deep learning use the 3-D CNN [33] for classification. Compared with other methods, the methods that use the 3-D CNN for hyperspectral classification not only can automatically learn the features from HSIs, but can also ensure the integrity of the information.

Given the advantages of the 3-D CNN in HSI processing, many related methods have been proposed successively. Wang *et al.* [34] introduced a fast spectral–spatial model with dense connectivity for hyperspectral classification. Zhong *et al.* [35] proposed a spectral–spatial network that is characterized by the residual structure to improve the hyperspectral classification performance effectively. Wang *et al.* [36] designed an alternately updated framework to learn the spectral–spatial features from HSIs. Fang *et al.* [37] presented a 3-D CNN that combined the dense connectivity with the spectralwise attention mechanism, which obtained satisfactory results. These methods bring about broad prospects for hyperspectral classification.

To solve the problem of limited labeled data in HSIs, various strategies have been proposed, such as the methods of data augmentation, which can generate new samples [38], [39], the methods of transfer learning, which can transfer samples from other data sets [40], [41], the methods of semisupervised learning, which can use labeled and unlabeled samples simultaneously [42], [43], the methods of the generative adversarial network (GAN), which can synthesize virtual samples [44], [45], and so on. Among these, the methods of data augmentation are widely used because of the relatively simple operation. Han *et al.* [46] proposed a strategy of data augmentation by rearranging the spatial positions of the pixels in each training sample for hyperspectral classification. Haut *et al.* [47] introduced a method for data augmentation that used random occlusion to increase the amount of labeled data. Wei and Wu [48] designed a pixel-pair method for data augmentation. Chen *et al.* [33] adopted the methods of changing radiation and mixing different samples to generate virtual data. These methods can achieve satisfactory results.

For the spectral–spatial classification methods used for hyperspectral classification, although the hyperspectral classification methods that use the 3-D CNN have excellent performance, the current methods focus only on increasing the depth of the network but ignore the disappearance of fine characteristics with an increase in the depth of the training process. With regard to the strategies of data augmentation, although various methods have been proposed, they ignore the imbalance between the numbers of hyperspectral samples in different categories, which leads to poor accuracy of a category with a small number of samples and limits the improvement of the AA. Although some articles remove the categories that have relatively small sample numbers to alleviate the problem of data imbalance, such as [39], [48], and [49], this approach will make the model lose the ability to classify these categories. To solve these problems, SSFRAN is proposed. In SSFRAN, the data balance augmentation can not only increase the amount of labeled data, but also balance the number of samples in different categories; the spectral–spatial residual module introduces the idea of residual connection [50] into a 3-D CNN, which can ease the problem of hyperspectral model degradation with lightweight parameters; the spectral–spatial fractal structure is inspired by the idea of fractal architecture [51], which can make the model wide and ensure the integrity of the extracted hyperspectral features; the spectral–spatial dimensional transformation can reduce the dimension of the hyperspectral feature maps whereas preserving the detailed information.

Specifically, the following are the main contributions.

1) The method of data balance augmentation is proposed. In this algorithm, different augmentation factors are assigned to different categories and the samples in each category are amplified by means of rotation, which can fix the problem of limited labeled data and imbalanced categories.
2) To make full use of the information contained in HSIs and ensure the integrity of fine features, different strategies are presented. Specifically, the spectral–spatial residual module is proposed. On the one hand, it can learn the spectral–spatial information from HSIs with lightweight parameters. On the other hand, it can avoid the problem of gradient disappearance as the model deepens. In addition, the spectral–spatial fractal structure is introduced. In this method, the spectral–spatial residual modules are embedded into a fractal structure that contains subpaths of different lengths, thus making the model wide and avoiding the loss of detailed information. Moreover, the spectral–spatial dimensional transformation module is proposed to reduce the number and size of the hyperspectral feature maps without losing the fine features.
3) There are many parameters in the 3-D CNN, and some could be prone to overfitting. Therefore, the optimization methods of BN, L2 regularization, and dropout are utilized in SSFRAN for optimization.
4) SSFRAN has a strong generalization capability, which was shown on three hyperspectral benchmark data sets, and it achieved the state-of-the-art results.

This article is organized as follows. In Section II, the basics of the 3-D CNN are presented. In Section III, the proposed methods are illustrated. In Section IV, the experiments and results are introduced. Finally, Section V draws a conclusion.

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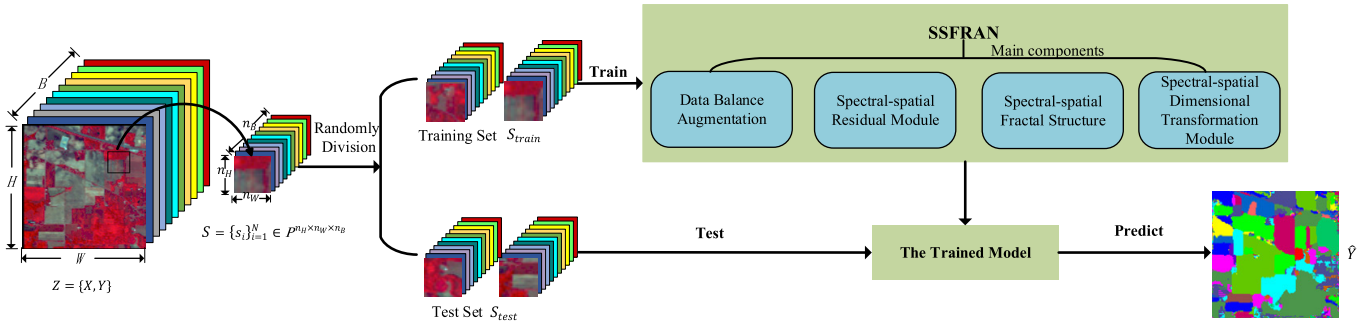


Fig. 1. Overview of the proposed method.

II. Basics of the 3-D CNN

The structure of the 3-D CNN is similar to that of the traditional 2-D CNN, except that the dimension of the input data is different. In this section, the operations of the 3-D convolution and 3-D average pooling, which are important components of the 3-D CNN are introduced concretely.

3-D convolution aims to obtain local information through the 3-D convolutional kernel acting on the input data. The value of $v_{l,j}^{x,y,z}$, which means the neuron of the lth layer at (x, y, z) on the jth feature map obtained by 3-D CNN, is expressed as follows:

$$v_{l,j}^{x,y,z} = \sum_{m} \sum_{p=0}^{P_l-1} \sum_{q=0}^{Q_l-1} \sum_{r=0}^{R_l-1} w_{l,j,m}^{p,q,r} v_{(l-1),m}^{(x+p),(y+q),(z+r)} + b_{l,j} \quad (1)$$

where m is the index of the feature maps in the $(l-1)$th layer. Additionally, P_l, Q_l, and R_l represent the three dimensions of height, width, and spectrum in the convolutional kernel. Additionally, the weight that connected with the neuron $v_{(l-1),m}^{(x+p),(y+q),(z+r)}$ in the lth layer at the position (p, q, r) on the mth feature map of the jth convolutional kernel is described as $w_{l,j,m}^{p,q,r}$. Furthermore, $b_{l,j}$ stands for the bias. After the operation of the 3-D convolution, the activation functions such as ReLU, Sigmoid, Tanh, are used to increase the nonlinearity of the network.

Pooling is used to reduce the dimension while maintaining the invariability of the features, thus preventing overfitting to a certain extent. 3-D pooling mainly includes 3-D maximum pooling and 3-D average pooling. The 3-D average pool is used throughout this article, and the equation is as follows:

$$P_{t,u,v} = \frac{1}{\left|y_{t,u,v}\right|} \sum_{i,j,k \in y_{t,u,v}} a_{i,j,k} \quad (2)$$

where $P_{t,u,v}$ stands for the value of the neuron at position (t, u, v) after the operation of 3-D average pooling. Here, $y_{t,u,v}$ represents the pooling area on the input. The value of the neuron in the pooling area $y_{t,u,v}$ is $a_{i,j,k}$, where i, j, and k index $a_{i,j,k}$ in the pooling area $y_{t,u,v}$.

III. Proposed Method

The overview of the proposed method is shown in Fig. 1. $Z = \{X, Y\}$ stands for the input of the model. $X \in \mathbb{R}^{H \times W \times B}$ is the data set of the HSIs, where H, W, and B refer to the three dimensions of height, width, and spectrum, respectively. Y is the label of the labeled pixel in the hyperspectral data set, $Y = \{y_i\}_{i=1}^{N}$, where N means the number of labeled pixels. To learn the spectral–spatial information from the HSIs adequately, the target pixel and its neighborhood pixels are generally regarded as the same category. Therefore, 3-D data cubes composed of the labeled center pixels and their adjacent pixels are used to generate the new sample set $S = \{s_i\}_{i=1}^{N} \in \mathbb{R}^{n_H \times n_W \times n_B}$, where n_H, n_W, and n_B are the dimensions of height, width, and spectrum in the 3-D data cube, respectively. In general, n_H and n_W are set to the same value. Then, the sample set S is randomly divided into the training set S_{train} and the test set S_{test} using the random sampling strategy. More specifically, the SSFRAN model is trained by using S_{train} to obtain the optimal model parameters in the training process. Afterward, the trained model is tested by S_{test}, and the predicted value $\hat{Y}$ can be finally obtained.

The structure of SSFRAN is introduced in this section. In SSFRAN, the data balance augmentation, which can address the problem of limited labeled data and imbalance categories in HSIs, the spectral–spatial residual module, which can acquire spectral–spatial features effectively, the spectral–spatial fractal structure, which can ensure the integrity of the information, and the spectral–spatial dimensional transformation module, which can reduce the size and number of feature maps, are presented in detail. Moreover, the framework of SSFRAN is described to further elaborate the design details of SSFRAN.

A. Data Balance Augmentation

The main idea of the data balance augmentation is to increase the sample number of each category according to the calculated augmentation factors through the transformation strategy in the training process. It can fix the problem of limited labeled data and unbalanced categories in the HSIs. The data balance augmentation mainly includes four steps: 1) count the number of samples in each category; 2) find out the maximum sample number of different categories; 3) calculate the augmentation factor for each class; and 4) increase the number of each category using the transformation strategy of rotation and interpolation according to the augmentation factor. The implementation details of the data balance augmentation are illustrated in Algorithm 1.

In the rotation operation, the center of the data cube is taken as the origin, the axes of x and y are defined as the rotation

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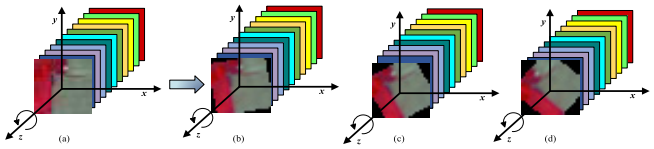


Fig. 2. Schematic of the transformation strategy in data balance augmentation. (a) Original data cube in the training set. (b) Data cube rotated by 10°. (c) Data cube rotated by 30°. (d) Data cube rotated by 50°.

plane, and the data cube is rotated counterclockwise along the z-axis. In addition, the operation of interpolation is adopted to keep the size of the data cube the same, which is equivalent to introducing noise and effectively increasing the diversity of the samples. Fig. 2 shows the details of the rotation operation.

Algorithm 1 Data Balance Augmentation

Input: The training set $S_{\text{train}} = \{s_i\}_{i=1}^{u}$, where u represents the number of training samples. The labels of the training set $Y_{\text{train}} = \{y_i\}_{i=1}^{u}$, and the number of categories is C in the hyperspectral data set.

Output: The new training set $S_{\text{train_aug}} = \{s_i'\}_{i=1}^{v}$ after the data balance augmentation, where v stands for the number of augmented training samples. The labels $Y_{\text{train_aug}} = \{y_i'\}_{i=1}^{v}$ for the new training set.

begin
  for $y_i \in Y_{\text{train}}$ **do**
    Count the number $T = \{t_i\}_{i=1}^{C}$ of samples in each category
  end
  for $t_i \in T$ **do**
    Compare the number of different categories to obtain the maximum $t_{\max} = \max(T) = \max\{t_i\}_{i=1}^{C}$
    Calculate the augmentation factor Φ for each category $\Phi = \{\varphi_i\}_{i=1}^{C} = \text{round}\left(\frac{t_{\max}}{T}\right) = \left\{\text{round}\left(\frac{t_{\max}}{t_i}\right)\right\}_{i=1}^{C}$
  end
  for $s_i \in S_{\text{train}}$, $y_i \in Y_{\text{train}}$ **do**
    for j from 1 to C **do**
      if $y_i = j$ **then**
        for k from 1 to φ_j **do**
          Rotate s_i by $10 \times (k-1)$ degrees and obtain s_{temp}
          Interpolate s_{temp} to the original sample size and obtain s_i'
          Get the corresponding label $y_i' = y_i$
        end
      end
    end
  end
end

B. Spectral–Spatial Residual Module

The spectral–spatial residual module which can take full advantage of the spectral–spatial information in the HSIs, is proposed. In the spectral–spatial residual module, the shortcut connection is introduced to ease the model degradation and gradient disappearance as the model deepens. In addition, inspired by the model of Inception V3 [52], the cascaded 3-D convolutions with small convolutional kernels are used to learn the spectral–spatial characteristics from the HSIs, which can reduce the parameter numbers effectively. Fig. 3 is the structure of the spectral–spatial residual module.

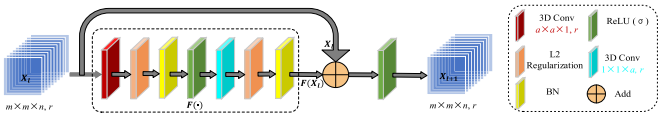


Fig. 3. Structure of the spectral–spatial residual module.

The input consists of r feature maps of size $m \times m \times n$ and is represented by X_l. Then, the value $F(X_l)$ obtained by the composite function $F(\cdot)$ is added in an elementwise manner to the input X_l by the shortcut connection. Finally, after the activation function ReLU, which is represented by σ, the output of the spectral–spatial residual module X_{l+1}, which contains r feature maps of size $m \times m \times n$, can be obtained. Additionally, X_{l+1} can be expressed by the following equation:

$$X_{l+1} = R(X_l) = \sigma[F(X_l) + X_l] \tag{3}$$

where $R(\cdot)$ stands for the function of the spectral–spatial residual module. In addition, the composite function is composed of the 3-D convolution using r kernels of size $a \times a \times 1$, which can extract spatial information, the operation of L2 regularization, which can ease overfitting, the BN, which can enhance the performance of the network, the activation function ReLU, which can enhance the nonlinear ability of the model, and the 3-D convolution adopting r kernels of size $1 \times 1 \times a$, which can extract the spectral information. Here, $\{a | a > 1, a \epsilon N^*\}$. Through the operation of factorizing the 3-D convolution of size $a \times a \times a$ into two cascaded 3-D convolutions of sizes $a \times a \times 1$ and $1 \times 1 \times a$, the parameter number of the convolutional kernel goes from a^3 to $a^2 + a$, which can not only obtain the spectral information and spatial information from the HSIs effectively but can also alleviate overfitting.

C. Spectral–Spatial Fractal Structure

The spectral–spatial fractal structure is mainly composed of spectral–spatial residual modules. It effectively increases the width of the proposed model and prevents the loss of detailed information by concatenating subpaths that contain various numbers of spectral–spatial residual modules. Fig. 4 is a diagram of the spectral–spatial fractal structure. The spectral–spatial fractal structure is composed of five subpaths, in which subpaths ① and ③ are composed of two spectral–spatial residual modules, whereas subpaths ②, ④, and ⑤ contain one spectral–spatial residual module. The parameter settings of each spectral–spatial residual module in the spectral–spatial fractal structure are the same. Consistent with the above, the function of the spectral–spatial residual module in the fractal structure is also represented by $R(\cdot)$. Suppose that the input X_f consists of t feature maps of size $p \times p \times q$. The features obtained after X_f passes through subpaths ① and ② are concatenated to obtain the intermediate output X_{temp}. X_{temp}

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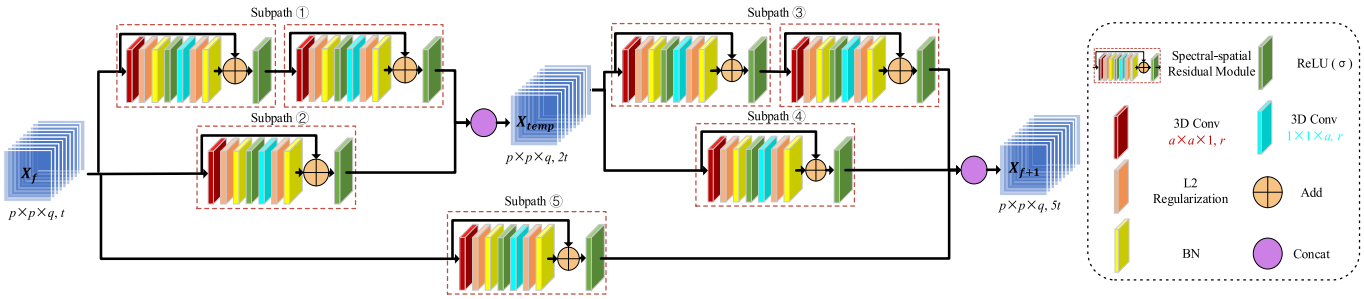


Fig. 4. Diagram of the spectral–spatial fractal structure.

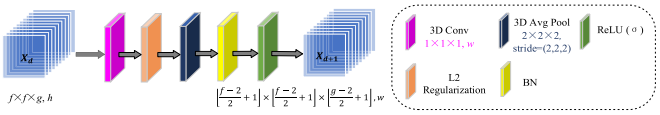


Fig. 5. Structure of the spectral–spatial dimensional transformation module.

consists of $2t$ feature maps of size $p \times p \times q$. Compared to X_f, X_{temp} contains twice as many feature maps as X_f, whereas the size of the feature map remains unchanged. It can be calculated by (4), as follows:

$$X_{\text{temp}} = R\left[R(X_f)\right] \cup R(X_f) \tag{4}$$

where $\cup$ represents the operation of concatenation. Afterward, the features obtained after X_{temp} passes through subpath ③, X_{temp} passes through subpath ④, and X_f passes through subpath ⑤ are concatenated to obtain the output X_{f+1}. X_{f+1} is composed of $5t$ feature maps of size $p \times p \times q$, and it can be calculated as follows:

$$X_{f+1} = R\left[R(X_{\text{temp}})\right] \cup R(X_{\text{temp}}) \cup R(X_f). \tag{5}$$

D. Spectral–Spatial Dimensional Transformation Module

The spectral–spatial dimensional transformation module can reduce the number and size of the hyperspectral feature maps without losing the fine features. In particular, it is parameter-efficient. The structure of the spectral–spatial dimensional transformation module is presented in Fig. 5.

We assume that the input X_d consists of h feature maps of size $f \times f \times g$. First, a 3-D convolution that adopts w kernels of size $1 \times 1 \times 1$ is utilized to decrease the number of feature maps in X_d. Here, $w < f$. Then, L2 regularization is adopted to ease overfitting. Then, the 3-D pooling with the filter of size $2 \times 2 \times 2$ and stride $(2, 2, 2)$ is used to decrease the size of the feature map. Next, BN is used to optimize the proposed model. Afterward, ReLU is used to increase the nonlinearity of SSFRAN. Finally, the output X_{d+1}, which contains w feature maps of size $\lfloor(f-2/2)+1\rfloor \times \lfloor(f-2/2)+1\rfloor \times \lfloor(g-2/2)+1\rfloor$, can be obtained. Here, the operation of rounding down is represented by the symbol $\lfloor\cdot\rfloor$.

E. Hyperspectral Classification Framework of SSFRAN

Taking SSFRAN on the IP data set as an example, the concrete structure of the proposed model is presented in Fig. 6. SSFRAN consists of the primary layer, the spectral–spatial fractal structure, the spectral–spatial dimensional transformation module, the global feature layer, and the output layer. In SSFRAN, all of the operations of the 3-D convolution adopt the "SAME" mode. To distinguish the 3-D convolution with different convolution kernel sizes adopted in SSFRAN, different colors are used in Fig. 6.

In SSFRAN, the 3-D image cube of size $15 \times 15 \times 200$ is chosen from the raw hyperspectral data set as a sample. First, the sample set is divided into training set and test set. Next, the number of training samples in the training set is increased by using the strategy of data balance augmentation. Then, after the sample passes through the primary layer, the feature map X_1 of size $8 \times 8 \times 100 \times 16$ can be obtained. The primary layer consists of a 3-D convolution that adopts 16 convolutional kernels of size $3 \times 3 \times 3$ and stride $(2, 2, 2)$, with L2 regularization, BN, and dropout. Then, the feature map X_2 of size $8 \times 8 \times 100 \times 80$ is obtained after X_1 passes through SSFS1. In the spectral–spatial fractal structure, spatial information and spectral information are extracted by using the 3-D convolution with the convolutional kernel of size $3 \times 3 \times 1$ and the 3-D convolution with the convolutional kernel of size $1 \times 1 \times 3$. Next, the feature map X_3 of size $4 \times 4 \times 50 \times 64$ can be obtained after X_2 passes through SSDTM1. Afterward, SSFS2 is adopted to learn the spectral–spatial information, obtaining the feature map X_4 of size $4 \times 4 \times 50 \times 320$. Then, SSDTM2 is utilized to decrease the dimension of X_4, obtaining X_5 of size $2 \times 2 \times 25 \times 256$. Then, the fine information of X_5 is extracted by SSFS3, to obtain the feature map X_6 of size $2 \times 2 \times 25 \times 1280$. Afterward, the feature map X_7 of size $1 \times 1 \times 12 \times 1024$ can be obtained through SSDTM3. Next, X_7 passes through the global feature layer, which consists of the 3-D convolution using the convolutional kernel of size $1 \times 1 \times 12$, L2 regularization, BN, the operation of flattening, and ReLU, to obtain the 1-D vector X_8 with the dimension of 2048. Finally, the output layer with the fully connected layer, L2 regularization, and dropout outputs the probabilities of the 16 categories.

In SSFRAN, the strategy of data balance augmentation is proposed in such a way that the limited labeled data

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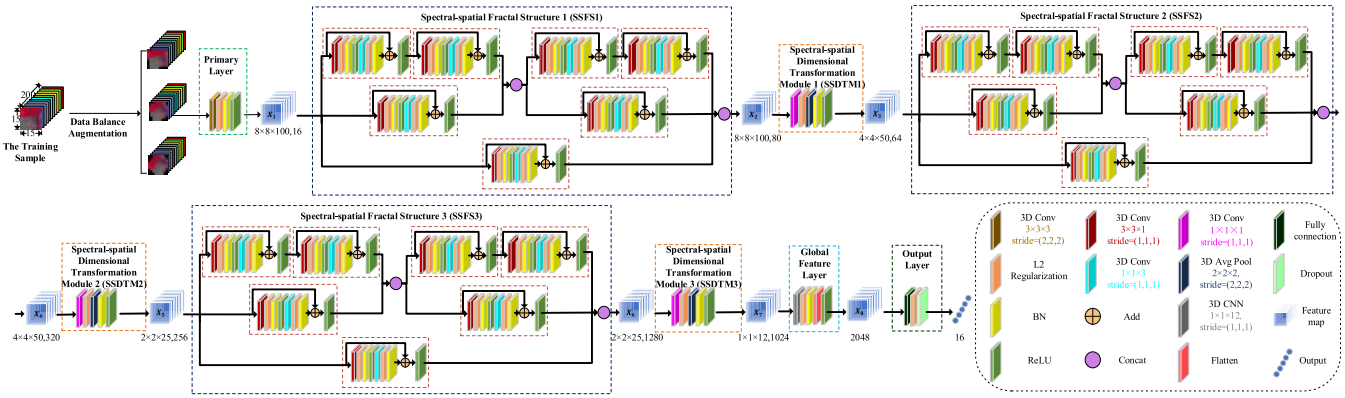


Fig. 6. Hyperspectral classification framework of SSFRAN.

can be amplified and the imbalanced categories can be balanced. In addition, the spectral–spatial residual module, which can adequately learn the spectral–spatial information and avoid the problem of gradient disappearance as the model deepens without lightweight parameters, is proposed. In addition, the spectral–spatial fractal structure is proposed, which contains multiple subpaths, thus making the model wider and protecting the detailed information. Furthermore, the spectral–spatial dimensional transformation module, which can decrease the number and size of the hyperspectral feature maps whereas preserving the detailed information effectively, is proposed. Moreover, BN, Dropout, and L2 regularization are used for optimization to ease overfitting and improve performance.

IV. Experiments and Results

The experimental data sets, experimental settings, and experimental performance evaluations are described here in detail.

A. Experimental Data Sets

To verify the performance of the SSFRAN, three benchmark data sets were utilized for the experiments.

The IP data set covers the scene of the IP test site and consists of 16 ground-truth classes. The wavelength ranges from 0.4×10^{-6} to 2.5×10^{-6} m, and the geometric resolution is 20 m. The original IP data set is composed of 145×145 pixels and 224 bands. The corrected IP data set consists of 200 bands.

The PU data set covers the scene of Pavia in Italy. The wavelength ranges from 0.43×10^{-6} to 0.86×10^{-6} m and the geometric resolution is 1.3 m. The IP data set contains nine ground categories. It is composed of 115 bands and 610×340 pixels. Additionally, there are 103 bands in the corrected PU data set.

The KSC data set covers the scene of the Kennedy Space Center in Florida and contains 13 ground-truth classes. The wavelength ranges from 0.4×10^{-6} to 2.5×10^{-6} m and the geometric resolution is 18 m. The original KSC data set has 512×614 pixels and 224 bands. After the noise bands have been removed, there are 176 bands in the corrected KSC data set.

B. Experimental Settings

In the experimental data sets, 20% of the samples were chosen at random for training and the remaining 80% of the samples were adopted for testing. Since the sample numbers in the different data sets are various, different batch sizes were set for the three data sets. The batch size of the IP data set, the PU data set, and the KSC data set was 32, 8, and 64, respectively. In addition, the number of training iterations was set to 50 000. The momentum optimizer and the exponential decay learning rate were utilized to make for rapid convergence of the model. Specifically, the value of the initial learning rate was set to 0.001, and the decay rate per 20 000 steps was 0.9. In addition, the probability of randomly dropping certain neurons in the strategy of the dropout was set to 0.5 to improve the performance of SSFRAN. Furthermore, the L2 regularization penalty ratio was set to 0.0001 in the L2 regularization.

The hardware environment of the experiment was a server with a Tesla V100 GPU and Xeon Gold 6139 CPU. In addition, the software environment of the experiment was the system of Ubuntu 18.04 with Python 3.6.9, TensorFlow 1.13.1, Keras 2.1.0, and Cuda 10.0.13.

C. Experimental Performance Evaluations

A series of evaluation indexes was used to evaluate SSFRAN. The OA refers to the proportion of correctly classified samples in the total samples. The AA refers to the ratio between the correctly classified sample numbers and the total sample numbers of each category. The Kappa coefficient is used to measure the similarity between the ground-truth distribution and the classification results. In theory, the closer these indicators used in this article are to 1, the better the classification effect will be.

In this section, SSFRAN is compared with other methods to highlight its superiorities. In addition, the importance of each module for SSFRAN is further illustrated by ablation studies.

1) Comparison With Different Methods: The classification results of SSFRAN on the IP data set, PU data set, and KSC data set are compared with different methods, as shown in Tables I–III, respectively. Here, the decision tree[1], naive

[1]http://scikit-learn.org

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TABLE I

CLASSIFICATION RESULTS ON THE IP DATA SET

No.	Class	Train/Test	Decision Tree	Naive Bayes	KNN	SVM	LDCR	S3EResBOF	MSDN	HybridSN	SSRN	SSFRAN
1	Asphalt	9/37	45.95	54.05	40.54	86.49	81.08	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
2	Corn-notill	285/1143	56.34	49.17	60.98	83.46	84.95	97.96	98.63	97.73	**100.00**	98.60
3	Corn-mintill	166/664	54.22	1.20	53.46	80.57	79.97	99.13	90.83	99.55	**99.85**	98.64
4	Corn	47/190	36.84	7.37	28.42	59.47	78.95	95.12	**100.00**	**100.00**	98.94	**100.00**
5	Grass-pasture	96/387	70.28	13.69	83.20	93.54	95.35	**100.00**	98.20	98.96	97.95	99.74
6	Grass-trees	146/584	84.08	70.21	97.09	95.38	96.06	98.66	98.78	99.32	99.48	**100.00**
7	Grass-pasture-mowed	5/23	56.52	39.13	82.61	91.30	78.26	75.00	**100.00**	**100.00**	**100.00**	**100.00**
8	Hay-windrowed	95/383	91.91	86.95	98.17	99.22	97.91	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
9	Oats	4/16	12.50	75.00	25.00	50.00	93.75	**100.00**	**100.00**	93.75	0.00	**100.00**
10	Soybean-notill	194/778	50.00	58.09	73.91	77.89	86.25	98.97	**99.72**	99.10	98.58	98.97
11	Soybean-mintill	491/1964	64.00	36.91	73.73	84.06	89.00	**99.64**	97.44	99.39	99.03	99.34
12	Soybean-clean	118/475	46.11	27.37	40.21	83.79	89.47	99.28	96.24	98.53	96.89	**99.37**
13	Wheat	41/164	93.90	91.46	96.34	97.56	**100.00**	**100.00**	99.38	**100.00**	**100.00**	**100.00**
14	Woods	253/1012	91.79	96.74	94.57	93.68	95.85	98.45	99.30	**99.90**	**99.90**	99.80
15	Building-Grass-Trees-Drives	77/309	44.66	24.59	22.01	62.78	67.96	98.91	97.73	**100.00**	**100.00**	98.38
16	Stone-Steel-Towers	18/75	76.00	86.67	82.67	86.67	92.00	95.65	97.14	91.89	**100.00**	94.67
OA (%)			65.38	48.73	71.53	85.13	88.70	98.68	97.74	99.13	**99.26**	**99.26**
AA (%)			60.94	51.16	65.81	82.87	87.93	98.84	98.34	98.63	93.17	**99.22**
Kappa (%)	Total	2045/8204	60.52	42.05	67.43	83.04	87.12	97.29	97.42	99.01	**99.15**	**99.15**
FLOPs ($\times 10^6$)			—	—	—	—	28.44	**6.37**	18.63	76.83	7.27	280.35
Test Time (s)			—	—	—	—	**1.79**	3.54	20.71	7.22	8.74	21.89

TABLE II

CLASSIFICATION RESULTS ON THE PU DATA SET

No.	Class	Train/Test	Decision Tree	Naive Bayes	KNN	SVM	LDCR	S3EResBOF	MSDN	HybridSN	SSRN	SSFRAN
1	Asphalt	1326/5305	85.01	64.13	89.01	94.89	95.89	99.83	99.94	99.94	**100.00**	99.89
2	Meadows	3729/14920	90.82	67.94	97.82	97.43	98.16	99.99	99.97	**100.00**	99.98	99.92
3	Gravel	419/1680	60.95	30.95	73.87	78.99	86.01	98.77	**99.94**	99.88	99.69	99.88
4	Trees	612/2452	86.46	93.31	84.54	95.68	97.02	99.44	**99.96**	99.51	**99.96**	99.39
5	Painted metal sheets	269/1076	99.16	99.44	99.16	99.44	99.81	99.79	**100.00**	**100.00**	**100.00**	**100.00**
6	Bare Soil	1005/4024	64.59	34.59	61.73	91.18	94.04	99.97	**100.00**	**100.00**	**100.00**	99.88
7	Bitumen	266/1064	69.17	89.94	84.96	85.99	90.88	98.62	**100.00**	**100.00**	**100.00**	99.91
8	Self-Blocking Bricks	736/2946	78.48	74.98	83.94	86.22	88.76	99.42	99.08	99.93	**99.97**	**99.97**
9	Shadows	189/758	**100.00**	98.94	99.87	**100.00**	**100.00**	99.84	**100.00**	99.34	99.87	**100.00**
OA (%)			83.79	66.39	88.58	94.07	95.70	99.69	99.89	99.93	**99.97**	99.88
AA (%)			81.63	72.69	86.10	92.20	94.51	99.76	99.88	99.85	**99.94**	99.87
Kappa (%)	Total	8551/34225	78.44	57.21	84.57	92.13	94.30	99.52	99.86	99.91	**99.96**	99.84
FLOPs ($\times 10^6$)			—	—	—	—	19.48	5.77	25.63	72.67	**4.17**	192.17
Test Time (s)			—	—	—	—	**5.35**	10.49	95.42	20.89	27.55	129.76

TABLE III

CLASSIFICATION RESULTS ON THE KSC DATA SET

NO.	Class	Train/Test	Decision Tree	Naive Bayes	KNN	SVM	LDCR	S3EResBOF	MSDN	HybridSN	SSRN	SSFRAN
1	Scrub	152/609	89.49	87.68	92.12	94.91	93.76	74.04	99.67	**100.00**	98.50	**100.00**
2	Willow swamp	48/195	76.92	67.69	78.97	82.56	79.49	73.43	**100.00**	90.72	92.23	98.97
3	CP hammock	51/205	81.46	78.05	82.93	81.95	92.68	78.29	97.56	96.09	**100.00**	98.54
4	Slash pine	50/202	46.53	42.08	52.97	60.89	65.84	87.04	90.95	88.61	54.59	**99.01**
5	Oak/Broadleaf	32/129	44.19	52.71	48.84	11.63	66.67	97.70	97.56	**98.45**	98.40	97.67
6	Hardwood	45/184	39.13	48.37	31.52	20.11	60.33	75.00	95.60	98.91	**100.00**	**100.00**
7	Swamp	21/84	55.95	86.90	84.52	79.77	96.43	0.00	**100.00**	88.09	**100.00**	**100.00**
8	Graminoid marsh	86/345	71.30	40.58	70.72	62.03	89.57	85.67	96.17	94.78	97.99	**99.42**
9	Spartina marsh	104/416	87.02	84.38	94.95	94.71	94.95	85.49	97.09	**100.00**	**100.00**	99.52
10	Cattail marsh	80/324	86.42	42.59	85.19	88.27	94.14	95.24	97.85	95.36	**100.00**	**100.00**
11	Salt marsh	83/336	94.35	92.86	89.58	94.94	97.62	99.67	**100.00**	**100.00**	**100.00**	99.40
12	Mud flats	100/403	83.62	76.92	77.67	82.63	91.07	**100.00**	99.50	99.25	**100.00**	**100.00**
13	Water	185/742	99.73	99.33	97.44	99.06	**100.00**	99.84	**100.00**	**100.00**	**100.00**	**100.00**
OA (%)			81.79	74.96	82.32	82.18	90.39	86.69	98.10	97.63	95.21	**99.62**
AA (%)			73.55	69.24	75.96	73.34	86.35	88.13	98.29	96.18	95.52	**99.43**
Kappa (%)	Total	1037/4174	79.72	72.13	80.29	80.08	89.30	80.88	97.84	97.35	94.66	**99.57**
FLOPs ($\times 10^6$)			—	—	—	—	26.33	**5.77**	31.04	72.68	6.49	265.63
Test Time (s)			—	—	—	—	**0.93**	1.19	29.53	2.97	15.88	10.79

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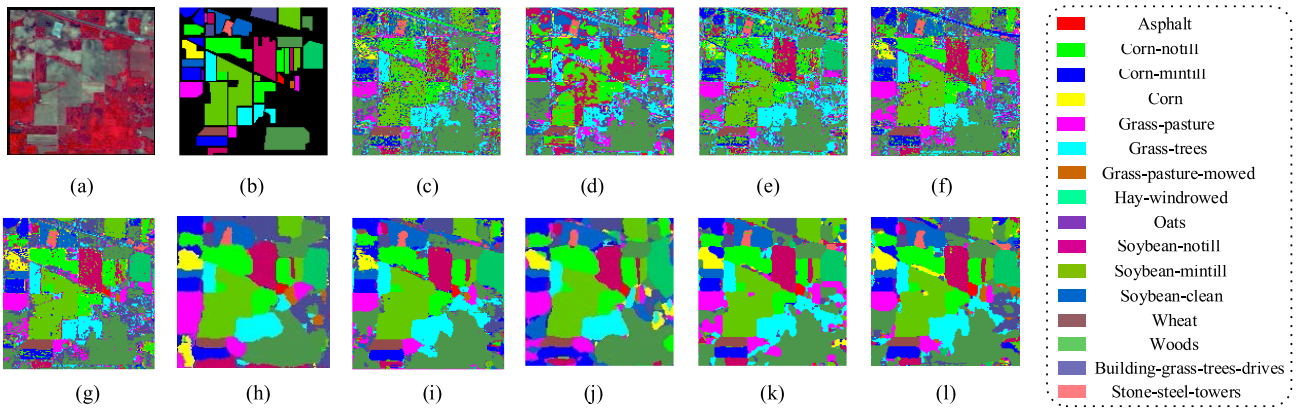


Fig. 7. Classification maps on the IP data set. (a) False color image. (b) Ground truth. (c) Decision Tree. (d) Naive Bayes. (e) KNN. (f) SVM. (g) LDCR. (h) S3EResBoF. (i) MSDN. (j) Hybrid. (k) SSRN. (l) SSFRAN.

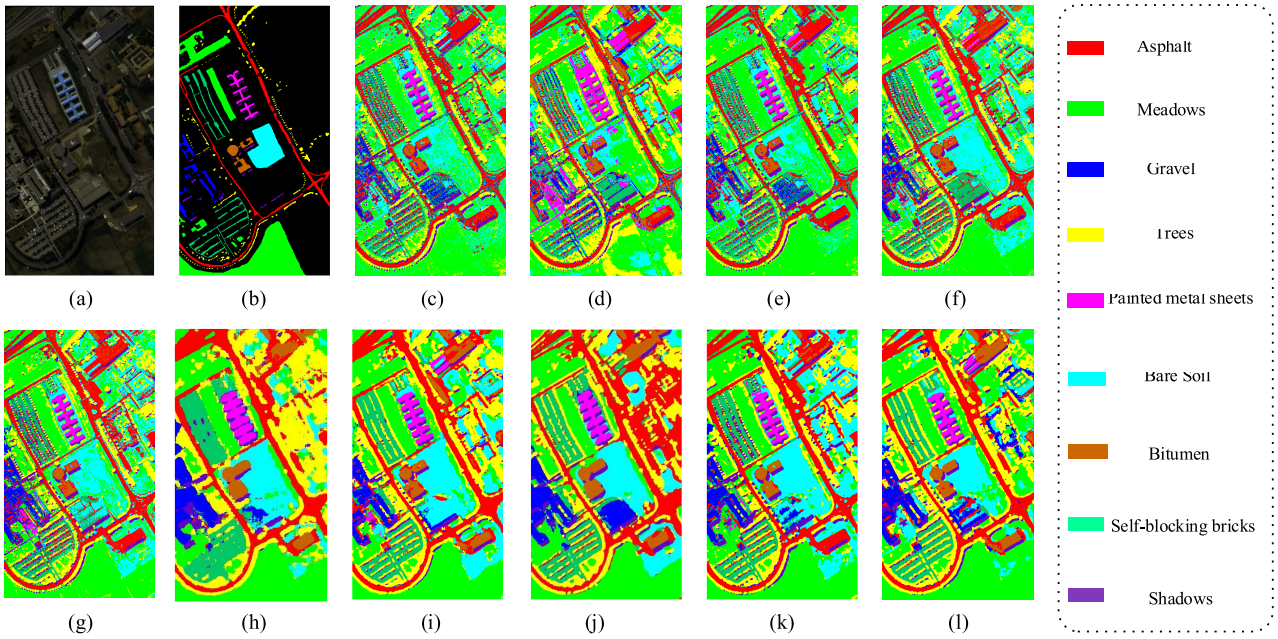


Fig. 8. Classification maps on the PU data set. (a) False color image. (b) Ground truth. (c) Decision Tree. (d) Naive Bayes. (e) KNN. (f) SVM. (g) LDCR. (h) S3EResBoF. (i) MSDN. (j) Hybrid. (k) SSRN. (l) SSFRAN.

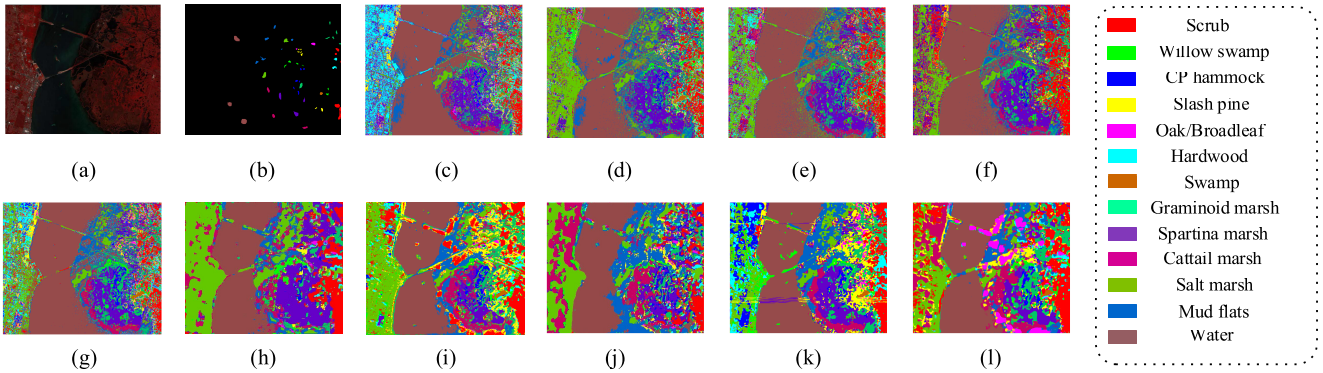


Fig. 9. Classification maps on the KSC data set. (a) False color image. (b) Ground truth. (c) Decision Tree. (d) Naive Bayes. (e) KNN. (f) SVM. (g) LDCR. (h) S3EResBoF. (i) MSDN. (j) Hybrid. (k) SSRN. (l) SSFRAN.

bayes[1], KNN[1], and SVM[1] all belong to traditional spectral-based classification methods, whereas LDCR [53] belongs to the spectral-based classification method using deep learning. In addition, S3EResBOF [54], MSDN [55], HybridSN [56], SSRN [35], and SSFRAN all belong to the spectral–spatial classification methods using deep learning. In different

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comparison methods, 20% of the samples were randomly chosen as the training set for fairness.

By comparing SSFRAN with the different methods, the following conclusions can be drawn.

1) SSFRAN has a strong ability to perform hyperspectral classification, especially for data sets with limited labeled data. SSFRAN reaches the state-of-the-art level. The OA, AA, and Kappa of SSFRAN are better than other methods on the IP data set and KSC data set, which contain a relatively small number of samples. In addition, on the PU data set, which contains abundant data, the evaluation indexes exceed 99%. This phenomenon is attributed to the strategy of data balance augmentation in SSFRAN, which can increase the number of labeled samples. With sufficient labeled data, the hyperspectral features can be fully learned and the classification accuracy can be improved. In addition, the spectral–spatial residual module, the spectral-fractal structure, and the spectral–spatial dimensional transformation module can retain the detailed information of HSIs and enhance the classification performance effectively.
2) In particular, SSFRAN has obvious advantages in the classification of categories with small sample numbers in different data sets. The accuracy of SSFRAN reached 100% on all the categories that contained the minimum number of samples, such as the oats in the IP data set, the shadows in the PU data set, and the swamp in the KSC data set, and the performance of SSFRAN in these cases far exceeded that of the other methods. This kind of situation is because of the strategy of data balance augmentation in SSFRAN which can balance the number of samples in different categories. This strategy can significantly increase the number of samples in the categories with small sample numbers so that the classification results of these categories can be significantly improved.
3) Compared with the traditional classification methods, the classification methods using deep learning showed the superiority for hyperspectral classification. This is because traditional classification methods generally adopt manual feature extraction, whereas the classification methods using deep learning can automatically learn high-level semantic information from images. In addition, the results show that the spectral–spatial classification methods have better performance than the spectral-based classification methods because they contain both spatial information and spectral information.
4) However, the proposed method still has some shortcomings. Since the spectral–spatial fractal structure of SSFRAN contains different subpaths, the integrity of information is guaranteed, but the structure of the model is relatively complex (compared via FLOPs) and the test time is relatively long. Therefore, how to reduce the complexity of the model and shorten the test time is still a problem worth studying. In addition, some articles point out that the random sampling strategy leads to overlapping between training and test samples [57], [58], which will produce optimistic results. How to solve this problem is also the direction of future research.

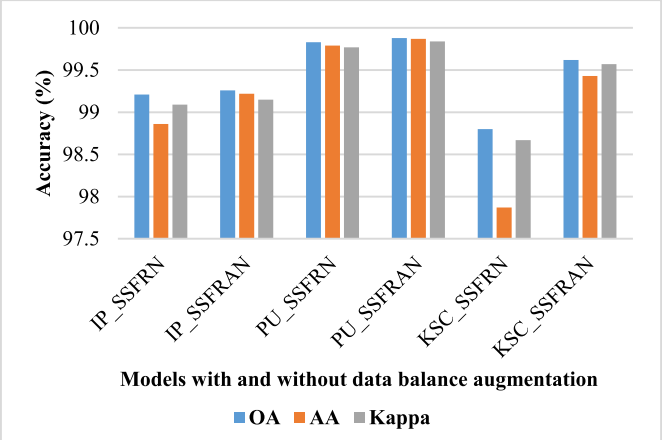


Fig. 10. Schematic of the ablation results about data balance augmentation on three data sets.

The classification maps of the various methods on the three data sets are shown in Figs. 7–9, respectively. The classification map of SSFRAN not only has clearer edges but also contains less noise compared with the other methods. This finding is attributed to the unique structure of SSFRAN, which makes the features of the HSIs be fully learned.

2) Effectiveness Analysis of Data Balance Augmentation: The OA, AA, and Kappa of the models with and without the data balanced augmentation are compared on the three data sets to prove the effectiveness of the data balanced augmentation. Fig. 10 shows the ablation results with regard to data balance augmentation on the experimental data sets.

As can be seen from Fig. 10, the OA, AA, and Kappa of the models with the data balance augmentation are higher than those of the models without the data balance augmentation on the three data sets. Additionally, the data balance augmentation had the most obvious effect on the KSC data set, which contains the least labeled data among the three experimental data sets. The above phenomenon proves that the data balance augmentation can solve the problem of limited labeled samples effectively. What is noteworthy is that the data balance augmentation can significantly improve the AA, which measures the mean of the different categories' accuracies, thus proving that the data balance augmentation can solve the problem of imbalanced categories effectively. In summary, the strategy of data balance augmentation can improve the classification performance of HSIs.

3) Effectiveness Analysis of the Spectral–Spatial Residual Module: The spectral–spatial residual module is mainly composed of the 3-D convolutions with the convolutional kernels of sizes $3 \times 3 \times 1$ and $1 \times 1 \times 3$, which can learn the spectral–spatial information from HSIs with lightweight parameters effectively. In addition, the shortcut connection is introduced into the spectral–spatial residual module, which is beneficial for information reuse and alleviates the problem of gradient

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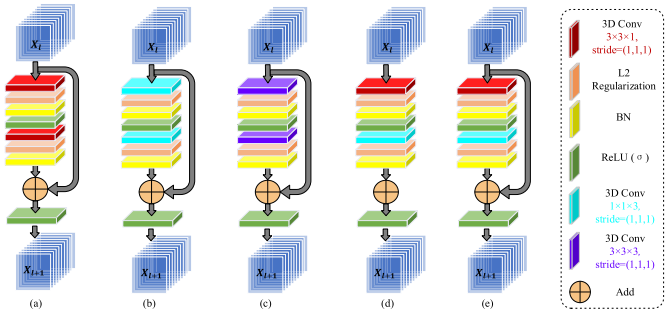


Fig. 11. Different ablation structures of the spectral–spatial residual module. (a) Spatial residual module. (b) Spectral residual module. (c) Spectral–spatial residual module adopting the 3-D convolution with large convolutional kernels. (d) Spectral–spatial module. (e) Spectral–spatial residual module.

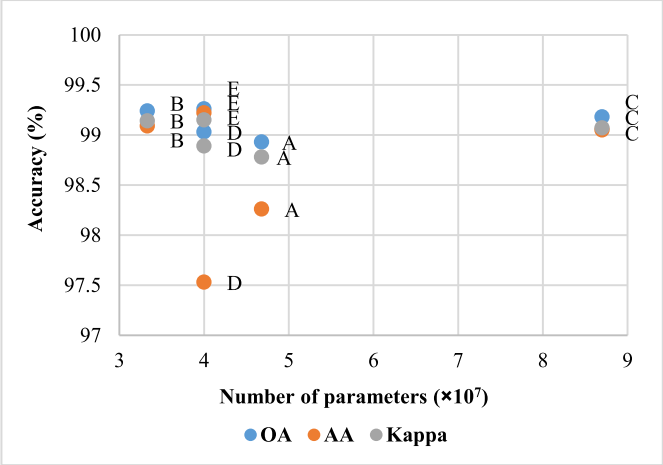


Fig. 12. Schematic of the ablation results about the spectral–spatial residual module on the IP data set.

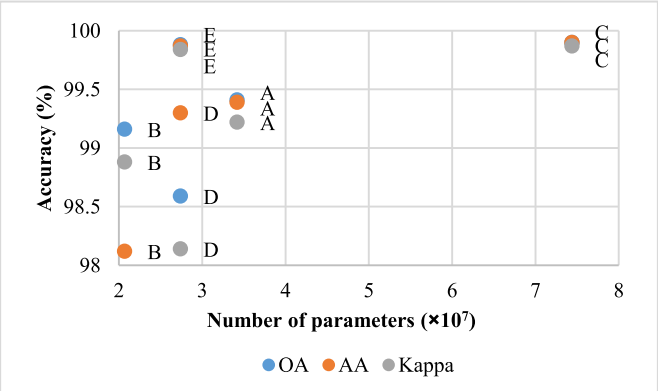


Fig. 13. Schematic of the ablation results about the spectral–spatial residual module on the PU data set.

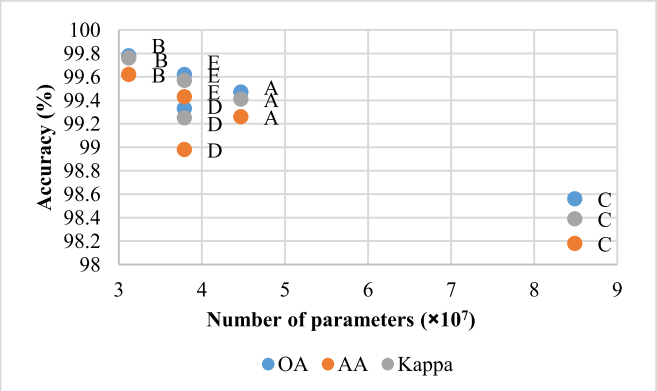


Fig. 14. Schematic of the ablation results about the spectral–spatial residual module on the KSC data set.

disappearance. Different ablation structures were adopted for comparison to prove the effectiveness of the spectral–spatial residual module, as shown in Fig. 11. The spatial residual module [Fig. 11(a)], which uses 3-D convolutions with the convolutional kernel of size $3 \times 3 \times 1$, the spectral residual [Fig. 11(b)], which uses 3-D convolutions with the convolutional kernel of size $1 \times 1 \times 3$, and the spectral–spatial residual module [Fig. 11(c)], which uses the 3-D convolutions with the convolutional kernel of size $3 \times 3 \times 3$, are adopted to verify the effectiveness of cascading the 3-D convolutions with small convolutional kernel sizes in the spectral–spatial residual module [Fig. 11(e)]. In addition, the spectral–spatial module [Fig. 11(d)] is included to prove the effectiveness of the shortcut connections in the spectral–spatial residual module [Fig. 11(e)].

Schematic diagrams of the ablation results about the spectral–spatial residual module on the IP data set, PU data set, and KSC data set are presented in Figs. 12–14, respectively. Here, A–E represent the networks using five ablation structures of the spectral–spatial residual module in Fig. 11, respectively. More specifically, E stands for SSFRAN; A represents the network that replaces the spectral–spatial residual module in SSFRAN with the spatial residual module; B represents the network that replaces the spectral–spatial residual module in SSFRAN with the spectral residual module; C represents the network that replaces 3-D convolutions that have small convolution kernel sizes in the spectral–spatial residual module of SSFRAN with the 3-D convolutions with large convolution kernel sizes; D represents the network that replaces the spectral–spatial residual module in SSFRAN with the spectral–spatial module.

From the view of the number of parameters, the parameter number of network C is the largest, followed by the parameter number of network A, and the number of parameters in network E and network D, which are equal, while the number of parameters is network B is the least. As far as OA, AA, and Kappa are concerned, network E has the best performance, followed by network B on the IP data set; on the KSC data set, network C has the best performance, followed by network E; on the PU data set, network B has the best performance, followed by network E. From the above analysis, it can be concluded that network E has the best performance when considering the number of parameters, which proves the effectiveness of the spectral–spatial residual module.

4) Effectiveness Analysis of the Spectral–Spatial Fractal Structure: The spectral–spatial fractal structure ensures the

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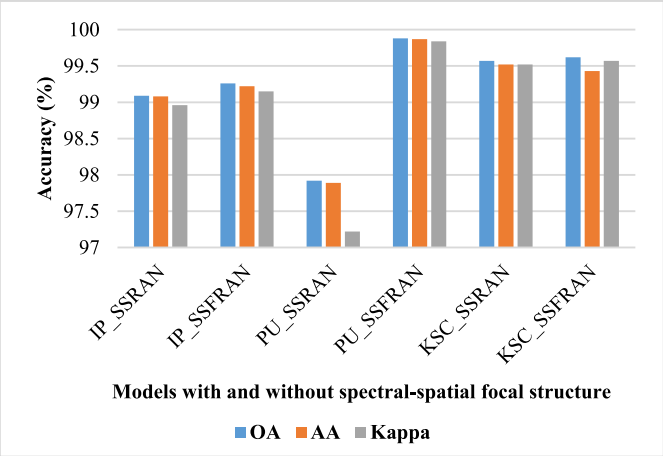


Fig. 15. Schematic of the ablation results about spectral–spatial focal structure on the three data sets.

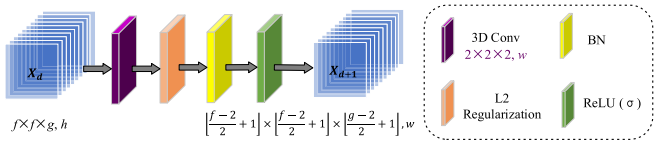


Fig. 16. Structure of the 3-D CNN layer used to replace the spectral–spatial dimensional transformation module in SSFRAN.

integrity of the information by concatenating subpaths that contain different numbers of spectral–spatial residual modules. To prove the validity of the spectral–spatial fractal structure, ablation studies about it were conducted on three data sets, and the result is shown in Fig. 15. Here, IP_SSRAN, PU_SSRAN, and KSC_SSRAN represent the networks that only use subpath ⑤ of the spectral–spatial fractal structure in SSFRAN on the data sets of IP, PU, and KSC, respectively. OA, AA, and Kappa of the networks that adopted the spectral–spatial fractal structure are overall higher than those of the networks without the spectral–spatial fractal structure on the three data sets, which proves the superiority of the spectral–spatial fractal structure.

5) Effectiveness Analysis of the Spectral–Spatial Dimensional Transformation Module: To verify the performance of the spectral–spatial dimensional transformation module, the 3-D convolutional layer is used to replace the spectral–spatial dimensional transformation module in SSFRAN, as shown in Fig. 16. The 3-D convolutional layer is composed the 3-D convolution with w kernels of size $2\times2\times2$, L2 regularization, BN, and ReLU. Ablation results on the IP, PU, and KSC data sets are shown in Figs. 17–19, respectively. Compared with networks using the 3-D convolutional layer, networks with the spectral–spatial dimensional transformation module have higher OA, AA, and Kappa, and fewer network parameters on three data sets. It can be concluded that the spectral–spatial dimensional transformation module can not only reduce the dimension of hyperspectral feature maps with fewer parameters, but also retain the detailed information.

6) Effectiveness Analysis of Optimization Methods: A series of optimization methods such as BN, Dropout, and L2 regularization, are utilized to improve the classification ability of SSFRAN. Ablation experiments that involved the different optimization methods were conducted to prove the effectiveness of the optimization methods. Fig. 20 is the schematic of the ablation results on the three data sets. It can be concluded that the OA, AA, and Kappa of the networks which adopted the optimization methods were usually higher than those in which one of the optimization methods was removed, which proves that BN, dropout, and L2 regularization can improve the classification ability on the HSIs.

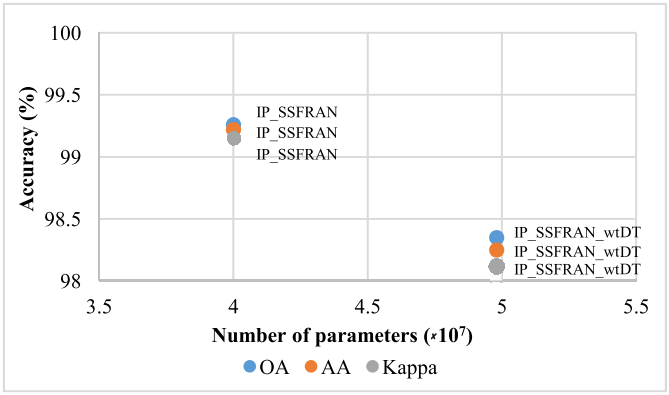


Fig. 17. Schematic of the ablation results about the dimensional transformation module on the IP data set.

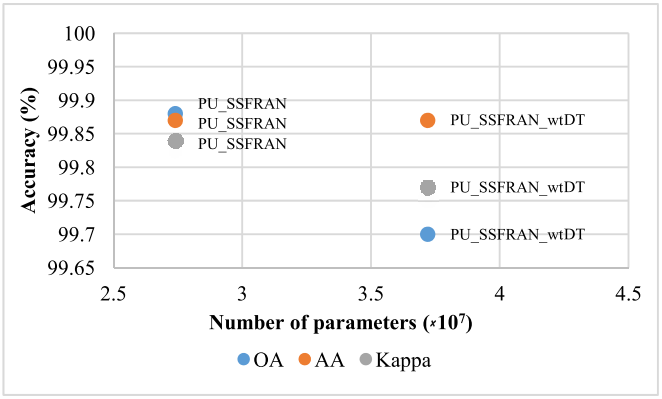


Fig. 18. Schematic of the ablation results about the dimensional transformation module on the PU data set.

7) Sensitivity to the Number of Training Samples: To explore the sensitivity of SSFRAN to the number of training samples, experiments were carried out on the experimental data sets. The corresponding results of different training sample numbers on the IP, PU, and KSC data sets are shown in Figs. 21–23, respectively. In general, when the proportion of training samples is 1%–20%, the OA, AA, and Kappa of SSFRAN gradually increased; and when the proportion of the training sample is more than 20%, the classification results gradually stabilized. Therefore, 20% training samples were selected for the training of SSFRAN. From the perspective of different training sample numbers, when the proportion of training samples is 1% or 3%, due to the random selection of samples, some categories of samples are not selected and the classification performance is not prominent; when

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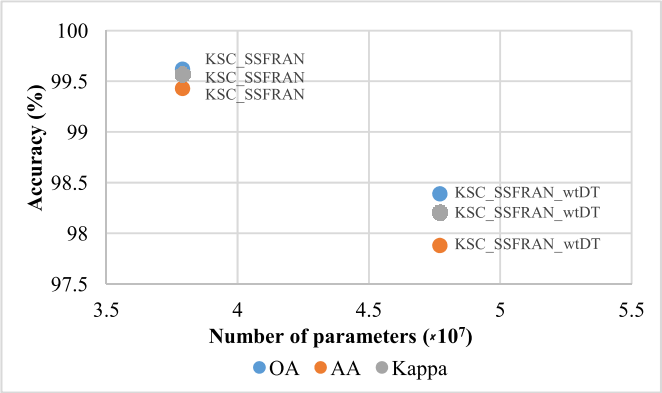


Fig. 19. Schematic of the ablation results about the dimensional transformation module on the KSC data set.

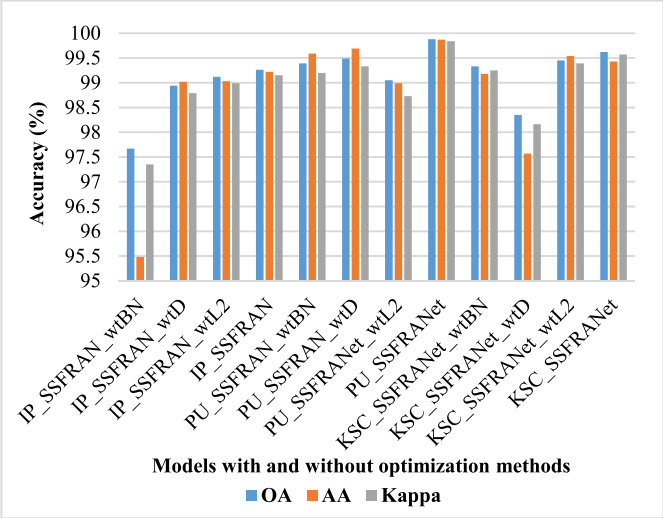


Fig. 20. Schematic of the ablation results about optimization methods on the three data sets.

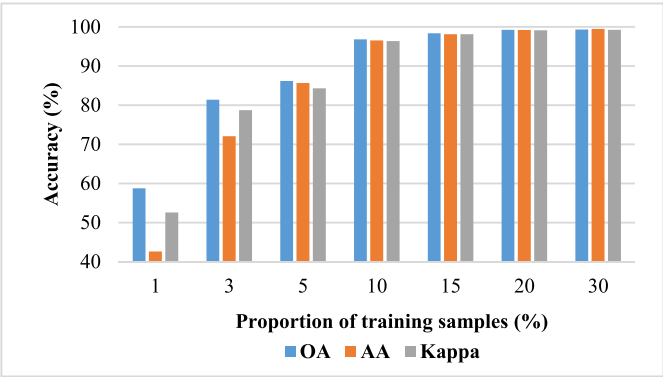


Fig. 21. Schematic of the results of different training sample numbers on the IP data set.

the proportion of training samples is 5%, the OA, AA and Kappa on the three data sets are all over 80%; and when the proportion of the training samples is 20%, the classification results all exceeded 99%, demonstrating the advantage of SSFRAN. From the perspective of different data sets, the IP and KSC data sets contain relatively few labeled samples,

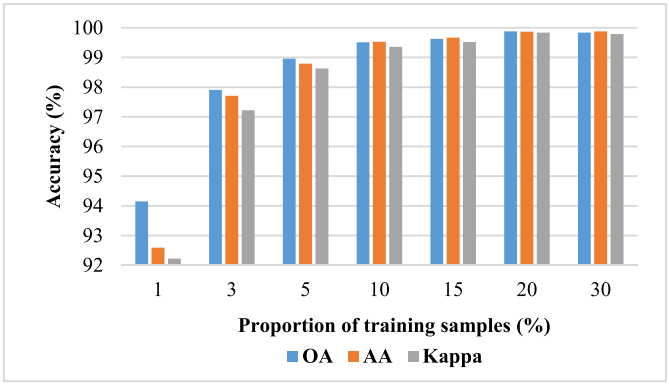


Fig. 22. Schematic of the results of different sample numbers on the PU data set.

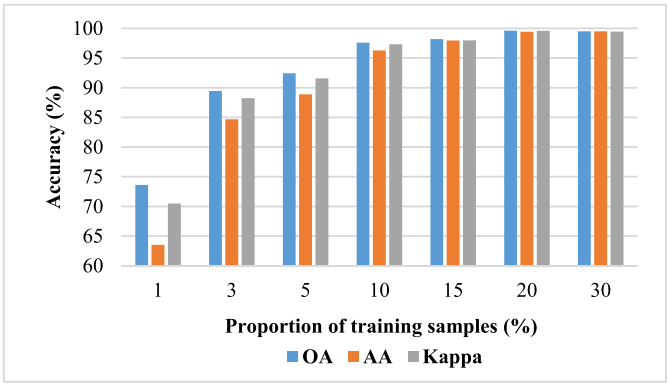


Fig. 23. Schematic of the results of different sample numbers on the KSC data set.

so the proportion of training samples has a relatively large impact on the classification results of these two data sets. However, the PU data set contains a large number of labeled samples, so it can still achieve satisfactory results without a large number of training samples.

V. Conclusion

In this article, an end-to-end 3-D CNN-based hyperspectral classification model—SSFRAN is introduced. In SSFRAN, the data balance augmentation is proposed to increase the number of labeled samples and to balance the sample numbers of different categories. In addition, the spectral–spatial residual module is introduced to learn the spectral–spatial features from the HSIs effectively and ease the gradient disappearance as the model deepens. Additionally, the spectral–spatial fractal structure is proposed to prevent the loss of detailed features. Furthermore, the spectral–spatial dimensional transformation module is presented effectively to decrease the dimension of the hyperspectral features whereas preserving the detailed information with lightweight parameters. All of the methods proposed above enable the performance of SSFRAN to reach the state-of-the-art level on experimental data sets.

The 3-D CNN-based hyperspectral classification methods have obtained satisfactory results, but this approach often requires enormous computational effort due to the many parameters. Therefore, how to reduce the computational effort

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of the 3-D CNN-based hyperspectral classification methods is our future research direction. Moreover, how to avoid the overlap of training and testing samples in HSIs is also worth studying.

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